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1 C:\Users\sumer\Desktop\Projects\Python\test\venv\Scripts\python
  .exe C:\Users\sumer\Desktop\Projects\Python\test\knn.py
2 Classification report for all hyperparameter combinations:
3 Hyperparameters: {'n_neighbors': 3, 'p': 1, 'weights': 'uniform
  '}
4 Mean test score: 0.771 (std: 0.003)
5 Classification report:
6           precision    recall  f1-score   support
7
8      0           0.83      0.93      0.88       7040
9      1           0.57      0.32      0.41       1960
10
11     accuracy                0.80       9000
12    macro avg           0.70      0.63      0.65       9000
13 weighted avg           0.77      0.80      0.78       9000
14
15 =====
16 =====
17 Hyperparameters: {'n_neighbors': 3, 'p': 1, 'weights': '
  distance'}
18 Mean test score: 0.768 (std: 0.004)
19 Classification report:
20           precision    recall  f1-score   support
21
22      0           0.83      0.93      0.88       7040
23      1           0.57      0.32      0.41       1960
24
25     accuracy                0.80       9000
26    macro avg           0.70      0.63      0.65       9000
27 weighted avg           0.77      0.80      0.78       9000
28
29 =====
30 =====
31 Hyperparameters: {'n_neighbors': 3, 'p': 2, 'weights': 'uniform
  '}
32 Mean test score: 0.773 (std: 0.007)
33 Classification report:
34           precision    recall  f1-score   support
35
36      0           0.83      0.93      0.88       7040
37      1           0.57      0.32      0.41       1960
38
39     accuracy                0.80       9000
40    macro avg           0.70      0.63      0.65       9000
41 weighted avg           0.77      0.80      0.78       9000
42
43 =====
44 =====

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42 Hyperparameters: {'n\_neighbors': 3, 'p': 2, 'weights': 'distance'}

43 Mean test score: 0.770 (std: 0.006)

44 Classification report:

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| 0            | 0.83      | 0.93   | 0.88     | 7040    |
| 1            | 0.57      | 0.32   | 0.41     | 1960    |
| accuracy     |           |        | 0.80     | 9000    |
| macro avg    | 0.70      | 0.63   | 0.65     | 9000    |
| weighted avg | 0.77      | 0.80   | 0.78     | 9000    |

53  
54 =====  
=====

55 Hyperparameters: {'n\_neighbors': 5, 'p': 1, 'weights': 'uniform'}

56 Mean test score: 0.788 (std: 0.004)

57 Classification report:

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| 0            | 0.83      | 0.93   | 0.88     | 7040    |
| 1            | 0.57      | 0.32   | 0.41     | 1960    |
| accuracy     |           |        | 0.80     | 9000    |
| macro avg    | 0.70      | 0.63   | 0.65     | 9000    |
| weighted avg | 0.77      | 0.80   | 0.78     | 9000    |

66  
67 =====  
=====

68 Hyperparameters: {'n\_neighbors': 5, 'p': 1, 'weights': 'distance'}

69 Mean test score: 0.785 (std: 0.002)

70 Classification report:

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| 0            | 0.83      | 0.93   | 0.88     | 7040    |
| 1            | 0.57      | 0.32   | 0.41     | 1960    |
| accuracy     |           |        | 0.80     | 9000    |
| macro avg    | 0.70      | 0.63   | 0.65     | 9000    |
| weighted avg | 0.77      | 0.80   | 0.78     | 9000    |

79  
80 =====  
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81 Hyperparameters: {'n\_neighbors': 5, 'p': 2, 'weights': 'uniform'}

82 Mean test score: 0.790 (std: 0.004)

83 Classification report:

|    |              | precision | recall | f1-score | support |
|----|--------------|-----------|--------|----------|---------|
| 86 | 0            | 0.83      | 0.93   | 0.88     | 7040    |
| 87 | 1            | 0.57      | 0.32   | 0.41     | 1960    |
| 88 |              |           |        |          |         |
| 89 | accuracy     |           |        | 0.80     | 9000    |
| 90 | macro avg    | 0.70      | 0.63   | 0.65     | 9000    |
| 91 | weighted avg | 0.77      | 0.80   | 0.78     | 9000    |

92

93 =====  
=====

94 Hyperparameters: {'n\_neighbors': 5, 'p': 2, 'weights': 'distance'}

95 Mean test score: 0.786 (std: 0.003)

96 Classification report:

|     |              | precision | recall | f1-score | support |
|-----|--------------|-----------|--------|----------|---------|
| 99  | 0            | 0.83      | 0.93   | 0.88     | 7040    |
| 100 | 1            | 0.57      | 0.32   | 0.41     | 1960    |
| 101 |              |           |        |          |         |
| 102 | accuracy     |           |        | 0.80     | 9000    |
| 103 | macro avg    | 0.70      | 0.63   | 0.65     | 9000    |
| 104 | weighted avg | 0.77      | 0.80   | 0.78     | 9000    |

105

106 =====  
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107 Hyperparameters: {'n\_neighbors': 7, 'p': 1, 'weights': 'uniform'}

108 Mean test score: 0.798 (std: 0.006)

109 Classification report:

|     |              | precision | recall | f1-score | support |
|-----|--------------|-----------|--------|----------|---------|
| 112 | 0            | 0.83      | 0.93   | 0.88     | 7040    |
| 113 | 1            | 0.57      | 0.32   | 0.41     | 1960    |
| 114 |              |           |        |          |         |
| 115 | accuracy     |           |        | 0.80     | 9000    |
| 116 | macro avg    | 0.70      | 0.63   | 0.65     | 9000    |
| 117 | weighted avg | 0.77      | 0.80   | 0.78     | 9000    |

118

119 =====  
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120 Hyperparameters: {'n\_neighbors': 7, 'p': 1, 'weights': 'distance'}

121 Mean test score: 0.794 (std: 0.005)

122 Classification report:

|  |  | precision | recall | f1-score | support |
|--|--|-----------|--------|----------|---------|
|--|--|-----------|--------|----------|---------|

124

|     |              |      |      |      |      |
|-----|--------------|------|------|------|------|
| 125 | 0            | 0.83 | 0.93 | 0.88 | 7040 |
| 126 | 1            | 0.57 | 0.32 | 0.41 | 1960 |
| 127 |              |      |      |      |      |
| 128 | accuracy     |      |      | 0.80 | 9000 |
| 129 | macro avg    | 0.70 | 0.63 | 0.65 | 9000 |
| 130 | weighted avg | 0.77 | 0.80 | 0.78 | 9000 |

131

132 =====  
=====

133 Hyperparameters: {'n\_neighbors': 7, 'p': 2, 'weights': 'uniform'}

134 Mean test score: 0.798 (std: 0.003)

135 Classification report:

|     |   |           |        |          |         |
|-----|---|-----------|--------|----------|---------|
| 136 |   | precision | recall | f1-score | support |
| 137 |   |           |        |          |         |
| 138 | 0 | 0.83      | 0.93   | 0.88     | 7040    |
| 139 | 1 | 0.57      | 0.32   | 0.41     | 1960    |

140

|     |              |      |      |      |      |
|-----|--------------|------|------|------|------|
| 141 | accuracy     |      |      | 0.80 | 9000 |
| 142 | macro avg    | 0.70 | 0.63 | 0.65 | 9000 |
| 143 | weighted avg | 0.77 | 0.80 | 0.78 | 9000 |

144

145 =====  
=====

146 Hyperparameters: {'n\_neighbors': 7, 'p': 2, 'weights': 'distance'}

147 Mean test score: 0.794 (std: 0.003)

148 Classification report:

|     |   |           |        |          |         |
|-----|---|-----------|--------|----------|---------|
| 149 |   | precision | recall | f1-score | support |
| 150 |   |           |        |          |         |
| 151 | 0 | 0.83      | 0.93   | 0.88     | 7040    |
| 152 | 1 | 0.57      | 0.32   | 0.41     | 1960    |

153

|     |              |      |      |      |      |
|-----|--------------|------|------|------|------|
| 154 | accuracy     |      |      | 0.80 | 9000 |
| 155 | macro avg    | 0.70 | 0.63 | 0.65 | 9000 |
| 156 | weighted avg | 0.77 | 0.80 | 0.78 | 9000 |

157

158 =====  
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159 Hyperparameters: {'n\_neighbors': 9, 'p': 1, 'weights': 'uniform'}

160 Mean test score: 0.804 (std: 0.003)

161 Classification report:

|     |   |           |        |          |         |
|-----|---|-----------|--------|----------|---------|
| 162 |   | precision | recall | f1-score | support |
| 163 |   |           |        |          |         |
| 164 | 0 | 0.83      | 0.93   | 0.88     | 7040    |
| 165 | 1 | 0.57      | 0.32   | 0.41     | 1960    |

166

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167 accuracy 0.80 9000
168 macro avg 0.70 0.63 0.65 9000
169 weighted avg 0.77 0.80 0.78 9000
170
171 =====
172 Hyperparameters: {'n_neighbors': 9, 'p': 1, 'weights': 'distance'}
173 Mean test score: 0.801 (std: 0.003)
174 Classification report:
175 precision recall f1-score support
176
177 0 0.83 0.93 0.88 7040
178 1 0.57 0.32 0.41 1960
179
180 accuracy 0.80 9000
181 macro avg 0.70 0.63 0.65 9000
182 weighted avg 0.77 0.80 0.78 9000
183
184 =====
185 Hyperparameters: {'n_neighbors': 9, 'p': 2, 'weights': 'uniform'}
186 Mean test score: 0.803 (std: 0.003)
187 Classification report:
188 precision recall f1-score support
189
190 0 0.83 0.93 0.88 7040
191 1 0.57 0.32 0.41 1960
192
193 accuracy 0.80 9000
194 macro avg 0.70 0.63 0.65 9000
195 weighted avg 0.77 0.80 0.78 9000
196
197 =====
198 Hyperparameters: {'n_neighbors': 9, 'p': 2, 'weights': 'distance'}
199 Mean test score: 0.801 (std: 0.004)
200 Classification report:
201 precision recall f1-score support
202
203 0 0.83 0.93 0.88 7040
204 1 0.57 0.32 0.41 1960
205
206 accuracy 0.80 9000
207 macro avg 0.70 0.63 0.65 9000
208 weighted avg 0.77 0.80 0.78 9000

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File - knn

209

210 =====  
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211 Best hyperparameters found: {'n\_neighbors': 9, 'p': 1, '  
weights': 'uniform'}

212 KNN Classification Accuracy: 0.7995555555555556

213

214 Process finished with exit code 0

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